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Inspection system and inspection method of building structures by using augmented reality based on building information modeling

Abstract

An inspection system of building structures includes a digital environment integration platform based on Common Data Environment, a physical inspection image corresponding to a structural part model in the Building Information Modeling, a server, and a mobile device. An inspection method of the building structures includes: connecting the server to the digital environment integration platform; recognizing feature data of the physical inspection image and sending the feature data to the server by the mobile device; acquiring the corresponding structural part model from the digital environment integration platform and sending the structural part model and a link of an inspection form to the mobile device by the server, wherein the inspection form complies with ISO 19650 standards; displaying an environment image and an augmented reality element of the structural part model on a screen by the mobile device; linking the mobile device to the inspection form.

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Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

(1) The present invention relates generally to an inspection of building structures, and more particularly to an inspection system and an inspection method of building structures by using augmented reality based on building information modeling.

Description of Related Art

(2) Before the construction of today's buildings, virtual three-dimensional models of the buildings, known as Building Information Modeling (BIM), are commonly created. To ensure the match between the building structures constructed and the Building Information Modeling, structural parts of the building structures (such as beams, columns, window openings, door openings, or other openings) have to be inspected throughout various stages of the construction process to recognize and rectify any errors in time.

(3) A conventional inspection method of the building structures involves an inspector carrying a laptop computer storing the Building Information Modeling and paper inspection forms to a location of the structural parts, which are about to be inspected, at the construction site. The inspector opens the Building Information Modeling in the laptop, searches for the structural parts, which are about to be inspected individually, in the Building Information Modeling, and compares the structural parts in the Building Information Modeling with actual structural parts at the construction site. After the comparison, the inspector records the results on the paper inspection forms.

(4) However, it is time-consuming for the inspector to search for a certain location among the extensive Building Information Modeling, causing delay in inspection operation and resulting in poor efficiency. In addition, recording numerous paper inspection forms at the construction site is prone to operational omissions or recording incorrectly.

(5) Therefore, the conventional inspection method of the building structures still has room for improvement.

BRIEF SUMMARY OF THE INVENTION

(6) In view of the above, the primary objective of the present invention is to provide an inspection system and an inspection method of building structures, which could effectively streamline the inspection process of the structural parts.

(7) The present invention provides an inspection system of building structures including a digital environment integration platform, at least one physical inspection image, a server, and a mobile device. The digital environment integration platform stores at least one structural part model of a building structure. The at least one physical inspection image is an image of the at least one structural part model. The server is connected to the digital environment integration platform. The server records a corresponding relationship between feature data of the at least one physical inspection image and the at least one structural part model. The corresponding relationship further includes a link of an inspection form. The mobile device has a camera module and a screen. The mobile device is connected to the server.

(8) The mobile device captures the at least one physical inspection image through the camera module and recognizes the feature data of the at least one physical inspection image. The mobile device sends the feature data recognized to the server.

(9) The server acquires the at least one structural part model from the digital environment integration platform based on the feature data and the corresponding relationship and sends the at least one structural part model and the link of the inspection form to the mobile device.

(10) The mobile device captures an environment image through the camera module and displays the environment image on the screen. The mobile device overlays the at least one structural part model acquired as an augmented reality element on the environment image displayed on the screen. The mobile device is linked to the inspection form through the link acquired to edit the inspection

form.
(11) The present invention provides an inspection method of building structures, including steps of: providing a digital environment integration platform storing at least one structural part model of a building structure; providing at least one physical inspection image, which is an image of the at least one structural part model; providing a server connected to the digital environment integration platform; recording, by the server, a corresponding relationship between feature data of the at least one physical inspection image and the at least one structural part model; the corresponding relationship further includes a link of an inspection form; connecting a mobile device to the server; capturing, by the mobile device, the at least one physical inspection image through the camera module and recognizing the feature data of the at least one physical inspection image by the mobile device; sending the feature data recognized to the server by the mobile device; acquiring the at least one structural part model from the digital environment integration platform by the server based on the feature data and the corresponding relationship and sending the at least one structural part model and the link of the inspection form to the mobile device by the server; capturing an environment image through the camera module by the mobile device and displaying the environment image on a screen of the mobile device by the mobile device; overlaying, by the mobile device, the at least one structural part model acquired as an augmented reality element in the environment image displayed on the screen; and linking the mobile device to the inspection form through the link acquired to edit the inspection form.
(12) With the aforementioned design, the user could perform the inspection of the structural parts in the building structures by simply carrying the mobile device and the physical inspection images, so that the problem that requires to search for the structural parts, which are about to be inspected one by one, in the Building Information Modeling or the problem that requires to bring along the paper inspection forms could be resolved. The inspection system and the inspection method of the building structures of the present invention could be more convenient for users to perform the inspection of the building structures, thereby effectively streamlining the inspection process of the structural parts.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) The present invention will be best understood by referring to the following detailed description of some illustrative embodiments in conjunction with the accompanying drawings, in which
(2) FIG. **1** is a schematic view of the inspection system of the building structures according to a first embodiment of the present invention;
(3) FIG. **2** is a flow chart of the inspection method of the building structures according to the first embodiment of the present invention;
(4) FIG. **3** is a schematic view of the structural part model according to the first embodiment of the present invention;
(5) FIG. **4** is a schematic view of the physical inspection image according to the first embodiment of the present invention;
(6) FIG. **5** is a schematic view of the floor plan according to the first embodiment of the present invention;
(7) FIG. **6** is a schematic view of the augmented reality element and the environment image displayed on the screen according to the first embodiment of the present invention;
(8) FIG. **7** is a schematic view of the field photo displayed on the screen according to the first embodiment of the present invention;
(9) FIG. **8** is another schematic view of the augmented reality element and the environment image displayed on the screen according to the first embodiment of the present invention; and

(10) FIG. **9** is a schematic view of the augmented reality element and the environment image displayed on a screen according to a second embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(11) An inspection system of building structures according to a first embodiment of the present invention is shown in FIG. **1** and includes a digital environment integration platform **10**, at least one physical inspection image **20**, a server **30**, and a mobile device **40**. The inspection system of the building structures is configured to perform an inspection method of building structures in the current embodiment, wherein the method includes steps S**11** to S**18** as shown in FIG. **2**. The inspection system of the building structures and the inspection method of the building structures would be described as below.

(12) The digital environment integration platform **10** is based on the Common Data Environment (CDE). The digital environment integration platform **10** could be, for example, a database or a cloud storage accessible for relevant participants to connect and access. The digital environment integration platform **10** stores at least one structural part model **50** of a building structure, wherein the at least one structural part model **50** corresponds to a structural part (such as a beam, a column, a window opening, a door opening, or other openings) of the building structure. In the current embodiment, the at least one structural part model **50** includes a plurality of structural part models **50**, wherein the plurality of structural part models **50** correspond to a plurality of structural parts. In practice, the digital environment integration platform **10** could be developed by using Unity.

(13) FIG. **3** shows one of the plurality of the structural part models **50**. Each of the plurality of structural part models **50** is one of a plurality of structural part models in the Building Information Modeling (BIM) of the building structure. In the current embodiment, at least one of the plurality of structural part models **50** defines a primary inspection portion **502** and a secondary inspection portion **504**. A color of the primary inspection portion **502** is different from a color of the secondary inspection portion **504**. For example, an opening shown in FIG. **3** is the primary inspection portion **502** having a color different from the other portions. The primary inspection portion **502** is a portion requiring tighter dimensional tolerances, such as the window opening, the door opening, or the other openings. The primary inspection portion **502** of actual structural parts needs subsequent installations of other elements, such as a door or a frame, so that the tighter dimensional tolerances are required. In practice, the Building Information Modeling could be developed by using Revit.

(14) In the current embodiment, the digital environment integration platform **10** further stores at least one field photo corresponding to at least one of the plurality of structural part models **50**. The at least one field photo could be an electronic file of a 360-degree photo captured by a 360-degree camera, but not limited thereto. The at least one field photo is a historical construction photo captured at a location of one of the plurality of structural parts, thereby allowing for subsequent reference. The at least one field photo could be a plurality of field photos respectively corresponding to the plurality of structural part models **50**. Each of the plurality of field photos is a historical construction photo in various stages captured at the location of each of the plurality of structural parts.

(15) In the current embodiment, the at least one physical inspection image **20** includes a plurality of physical inspection images **20**. Each of the plurality of physical inspection images **20** is printed on a sheet of paper **52** and corresponds to one of the plurality of structural part models **50** in the Building Information Modeling. FIG. **4** shows one of the physical inspection images **20** that corresponds to the structural part model **50** shown in FIG. **3**. Personnel prints at least one portion of the structural part model **50** on the paper **52**. In addition, location data **54** of the structural part model **50** on a floor plan **53** (shown in FIG. **5**) is also printed on the paper **52**. The floor plan **53** shown in FIG. **5** includes a plurality of inspection points 1 to 3, wherein the plurality of inspection points 1 to 3 respectively have one corresponding structural part model **50**. Taking the structural part model **50** of the inspection point 1 as an example, the location data **54** may include grid (line) location “Line: F/5” and information indicating the location of the structural part model **50**, such as

a construction site name “Company OO Plant 1”, a building name and a floor “Building B 2F”, and an inspection point name “Inspection Point 1”. The structural part model **50** and the physical inspection image **20** are further marked with dimensions associated with the primary inspection portion **502** for user reference.

(16) The server **30** is connected to the digital environment integration platform **10** and is configured to access data in the digital environment integration platform **10**. In practice, the digital environment integration platform **10** could also be integrated with the server **30** in a host. In practice, the server **30** could be developed by using GitHub. The server **30** records a corresponding relationship between feature data of each of the plurality of physical inspection images **20** and each of the plurality of structural part models **50**. The corresponding relationship further includes a link of an inspection form corresponding to each of the plurality of structural part models **50**. Each of the inspection forms is in an electronic file stored in the cloud storage; each of the links is configured to be linked to the inspection form in the cloud storage, but not limited thereto; each of the inspection forms could also be stored in the digital environment integration platform **10**. Each of the inspection forms complies with ISO 19650 standards. In practice, the feature data of the plurality of physical inspection images **20** could be created by using Vuforia, which uses the technology of Marker Augmented Reality (Marker AR).

(17) The mobile device **40** is connected to the server **30**. In the current embodiment, the mobile device **40** could be a device convenient for a user (such as an inspector) to carry and connect to the server **30** through the Internet, such as a smartphone, a tablet, a notebook, etc. The mobile device **40** has a screen **42**, a camera module **44**, and a positioning module **46**. The screen **42** could be a touch screen as an example.

(18) The mobile device **40** executes an application software to be connected to the server **30**, and then the user inputs an account and a password in the mobile device **40** to log in the server **30**. The positioning module **46** of the mobile device **40** is configured to acquire positioning data. For example, the positioning module **46** may acquire the positioning data through an indoor positioning device (not shown) disposed at a current floor of a construction site of the building structure; the indoor positioning device could position by using wireless signals, such as WIFI or Bluetooth; the mobile device **40** acquires the positioning data based on the wireless signals received by the positioning module **46**.

(19) The user takes out one of the plurality of paper physical inspection images **20** ready for use when walking to the inspection point, which is about to be inspected, of the construction site of the building structure.

(20) Afterward, the user operates the mobile device **40** to scan the physical inspection image **20**, that is, the mobile device **40** captures the physical inspection image **20** through the camera module **44** and recognizes the feature data thereof. The mobile device **40** sends the feature data to the server **30**.

(21) The server **30** acquires corresponding one of the structural part models **50** from the digital environment integration platform **10** based on the feature data and the corresponding relationship. The server **30** sends the corresponding structural part model **50** and the corresponding link of the inspection form to the mobile device **40**.

(22) As shown in FIG. **6**, the mobile device **40** continuously captures an environment image **56** of the construction site through the camera module **44** and displays the environment image **56** on the screen **42** of the mobile device **40**. The mobile device **40** overlays the structural part model **50** acquired as an augmented reality element **58** on the environment image **56** displayed on the screen **42**.

(23) In the current embodiment, the corresponding relationship recorded by the server **30** further includes the location data **54** corresponding to each of the plurality of structural part models **50**. In the step of sending the structural part model **50** and the corresponding link of the inspection form to the mobile device **40** by the server **30**, the server **30** further sends the location data **54**

corresponding to the structural part model **50** to the mobile device **40**; the mobile device **40** displays the location data **54** on the screen **42**.
(24) In the current embodiment, when the mobile device **40** displays the augmented reality element **58** on the screen **42**, the primary inspection portion **502** and the secondary inspection portion **504** are displayed simultaneously. The color of the primary inspection portion **502** is different from the color of the secondary inspection portion **504**. As shown in FIG. **6**, an opening of the augmented reality element **58** is displayed in a deeper color to remind the user to pay attention to the primary inspection portion **502** and to compare the primary inspection portion **502** with a primary inspection portion (such as the opening) of a structural part **64** in detail. Preferably, the primary inspection portion **502** of the augmented reality element **58** could be marked with dimensions associated with the primary inspection portion **502** shown in FIG. **3** and FIG. **4**, so that the user could compare the actual dimensions of the primary inspection portion of the structural part **64** measured with a tape measure at the construction site with the dimensions marked in the primary inspection portion **502** of the augmented reality element **58**.
(25) In the current embodiment, the mobile device **40** displays a manipulation interface **60** on the screen **42**; the manipulation interface **60** includes a plurality of manipulation icons **62***a* to **62***e*; the manipulation icon **62***a* is configured to store the environment image **56** as a static image file and to store the environment image **56** overlaid with the augmented reality element **58** as another static image file; the manipulation icon **62***b* is configured to display the at least one field photo; the manipulation icon **62***c* is configured to be linked to the inspection form; the manipulation icon **62***d* is configured to rotate the augmented reality element **58**; the manipulation icon **62***e* is configured to pan the augmented reality element **58**. Additionally, other manipulation icons could be included to zoom in and out of the augmented reality element **58**. The user could rotate, pan, zoom in, and zoom out the augmented reality element **58** with fingers on the screen **42**, so that the user could compare the augmented reality element **58** with the structural part **64** in the environment image **56**.
(26) When the user selects the manipulation icon **62***b*, the mobile device **40** sends a request command to the server **30**, wherein the request command is a request to acquire the at least one field photo. The server **30** acquires the at least one field photo from the digital environment integration platform **10** based on the request command and sends the at least one field photo to the mobile device **40**. As shown in FIG. **7**, the mobile device **40** displays the at least one field photo **66** on the screen **42**, so that the user could manipulate the 360-degree field photo **66** on the screen **42** with fingers to change viewing angles, thereby facilitating comparison of the field photo **66** with the current situations at the construction site.
(27) After the user selects the manipulation icon **62***c*, the mobile device **40** is linked to the inspection form corresponding to the structural part model **50** through the link acquired to edit the inspection form.
(28) In this way, the user could compare the augmented reality element **58** with the structural part **64** in the environment image **56** to determine whether the structural part **64** in the environment image **56** matches or is consistent with the structural part model **50**, and the results of the inspection could be filled in the inspection form through the mobile device **40**. The mobile device **40** could further automatically append the image file with the environment image **56** and the another image file with the augmented reality element **58** and the environment image **56** to the inspection form.
(29) Additionally, in the current embodiment, user data of the user, such as a user name, an employee number, etc., could be inputted into the mobile device **40**. When the inspection form is edited through the mobile device **40**, the mobile device **40** automatically inputs the user data into the inspection form, thereby ensuring that the inspection form is filled out by the user.
(30) In this way, the user could perform the inspection of structural parts in building structures by simply carrying the mobile device **40** and the physical inspection images **20**, thereby resolving the problem that requires to search for the structural parts, which are about to be inspected one by one,

in the Building Information Modeling or the problem that requires to bring along the paper inspection forms.

(31) Optionally, when the mobile device **40** detects a mismatch between a shape of the structural part **64** in the environment image **56** and a shape of the augmented reality element **58** of the structural part model **50**, the mobile device **40** generates a prompt message. The prompt message could be displayed on the screen **42**, for example showing “Mismatch detected”.

(32) Optionally, when the mobile device **40** detects a match between the positioning data and the location data **54**, the mobile device **40** displays the augmented reality element **58** (shown in FIG. **6**) on the screen **42**. When the mobile device **40** detects a mismatch between the positioning data and the location data **54**, the mobile device **40** generates a prompt message **68**. For example, as shown in FIG. **8**, the prompt message **68**, “Model does not match current location”, is displayed on the screen **42** to remind the user that the current location is incorrect. In practice, when the mobile device **40** detects a mismatch between the positioning data and the location data **54**, the mobile device **40** does not display and/or hide the augmented reality element **58** on the screen **42** but displays the prompt message **68** directly. When the user reaches the correct location and the mobile device **40** detects a match between the positioning data and the location data **54**, the augmented reality element **58** is displayed and the prompt message **68** is canceled.

(33) As shown in FIG. **9**, an inspection system and an inspection method of building structures according to a second embodiment of the present invention are almost the same as that of the first embodiment, except that when the mobile device **40** displays the augmented reality element **58** on the screen **42**, the mobile device **40** further displays a manipulation icon **70** on the screen **42**; the manipulation icon **70** is a slider configured to adjust a transparency of the augmented reality element **58**. When the user manipulates the manipulation icon **70** on the screen **42**, the mobile device **40** adjusts the transparency of at least one portion of the augmented reality element **58**. For example, the mobile device **40** could adjust the transparency of the entire augmented reality element **58**, so that the user could compare the structural part **64** in the environment image **56** with the augmented reality element **58**, which is adjusted to be semi-transparent, through the semi-transparent augmented reality element **58** on the screen **42**.

(34) Alternatively, when the user manipulates the manipulation icon **70** on the screen **42**, the mobile device **40** adjusts the transparency of the augmented reality element **58**; for example, the mobile device **40** adjusts the primary inspection portion **502** of the augmented reality element **58** to be semi-transparent. The mobile device **40** compares the semi-transparent primary inspection portion **502** of the augmented reality element **58** with a portion of the structural part **64** in the environment image **56**, that is, the mobile device **40** compares the semi-transparent primary inspection portion **502** with the actual primary inspection portion of the structural part **64**; when the semi-transparent primary inspection portion **502** does not match the actual primary inspection portion of the structural part **64**, the mobile device **40** generates a prompt message. The prompt message could be displayed on the screen **42**, for example showing “Mismatch detected”.

(35) With the aforementioned design, the inspection system and the inspection method of the building structures of the present invention could be more convenient for users to perform the inspection of the building structures, thereby effectively streamlining the inspection process of the structural parts.

(36) It must be pointed out that the embodiments described above are only some preferred embodiments of the present invention. All equivalent structures which employ the concepts disclosed in this specification and the appended claims should fall within the scope of the present invention.

Claims

1. An inspection system of building structures, comprising: a digital environment integration platform storing at least one structural part model of a building structure; at least one physical inspection image, which is an image of the at least one structural part model; a server connected to the digital environment integration platform, the server recording a corresponding relationship between feature data of the at least one physical inspection image and the at least one structural part model, the corresponding relationship further including a link of an inspection form; and a mobile device having a camera module and a screen, the mobile device connected to the server; wherein the mobile device captures the at least one physical inspection image through the camera module and recognizes the feature data of the at least one physical inspection image; the mobile device sends the feature data recognized to the server; wherein the server acquires the at least one structural part model from the digital environment integration platform based on the feature data and the corresponding relationship and sends the at least one structural part model and the link of the inspection form to the mobile device; wherein the mobile device captures an environment image through the camera module and displays the environment image on the screen; the mobile device overlays the at least one structural part model acquired as an augmented reality element on the environment image displayed on the screen; the mobile device is linked to the inspection form through the link acquired to edit the inspection form.
2. The inspection system of the building structures as claimed in claim 1, wherein the corresponding relationship further includes location data corresponding to the at least one structural part model; the mobile device has a positioning module configured to acquire positioning data; the server further sends the location data to the mobile device; when the mobile device detects a match between the positioning data and the location data, the mobile device displays the augmented reality element on the screen.
3. The inspection system of the building structures as claimed in claim 2, wherein when the mobile device detects a mismatch between the positioning data and the location data, the mobile device generates a prompt message.
4. The inspection system of the building structures as claimed in claim 2, wherein when the mobile device detects a mismatch between the positioning data and the location data, the mobile device does not display the augmented reality element on the screen.
5. The inspection system of the building structures as claimed in claim 1, wherein the digital environment integration platform includes at least one field photo corresponding to the at least one structural part model; the mobile device sends a request command to the server, wherein the request command is a request to acquire the at least one field photo; the server acquires the at least one field photo from the digital environment integration platform based on the request command and sends the at least one field photo to the mobile device; the mobile device displays the at least one field photo on the screen.
6. The inspection system of building structures as claimed in claim 1, wherein the mobile device has user data; when the mobile device edits the inspection form, the user data is automatically inputted into the inspection form.
7. The inspection system of the building structures as claimed in claim 1, wherein when the mobile device detects a mismatch between a shape of a structural part in the environment image and a shape of the augmented reality element of the at least one structural part model, the mobile device generates a prompt message.
8. The inspection system of the building structures as claimed in claim 1, wherein the at least one structural part model defines a primary inspection portion and a secondary inspection portion; a color of the primary inspection portion is different from a color of the secondary inspection portion; when the mobile device displays the augmented reality element on the screen, the primary inspection portion and the secondary inspection portion are displayed simultaneously.
9. The inspection system of the building structures as claimed in claim 1, wherein the at least one

structural part model defines a primary inspection portion and a secondary inspection portion; when the mobile device displays the augmented reality element on the screen, the mobile device adjusts the primary inspection portion on the augmented reality element to be semi-transparent; the mobile device compares the semi-transparent primary inspection portion on the augmented reality element with a portion of a structural part in the environment image; the mobile device generates a prompt message when a mismatch between the semi-transparent primary inspection portion on the augmented reality element and the portion of the structural part in the environment image is detected.

10. The inspection system of the building structures as claimed in claim 1, wherein when the mobile device displays the augmented reality element on the screen, the mobile device adjusts a transparency of at least one portion of the augmented reality element.

11. An inspection method of building structures, comprising steps of: providing a digital environment integration platform storing at least one structural part model of a building structure; providing at least one physical inspection image, which is an image of the at least one structural part model; providing a server connected to the digital environment integration platform; recording, by the server, a corresponding relationship between feature data of the at least one physical inspection image and the at least one structural part model; the corresponding relationship further including a link of an inspection form; connecting a mobile device to the server; capturing, by the mobile device, the at least one physical inspection image through a camera module and recognizing, by the mobile device, the feature data of the at least one physical inspection image; sending the feature data recognized to the server by the mobile device; acquiring the at least one structural part model from the digital environment integration platform by the server based on the feature data and the corresponding relationship and sending the at least one structural part model and the link of the inspection form to the mobile device by the server; capturing an environment image through the camera module by the mobile device and displaying the environment image on a screen of the mobile device by the mobile device; overlaying, by the mobile device, the at least one structural part model acquired as an augmented reality element on the environment image displayed on the screen; and linking the mobile device to the inspection form through the link acquired to edit the inspection form.

12. The inspection method of the building structures as claimed in claim 11, wherein the corresponding relationship further includes location data corresponding to the at least one structural part model; the mobile device has a positioning module configured to acquire positioning data; the inspection method of the building structures further comprises: sending the location data to the mobile device by the server; displaying the augmented reality element on the screen by the mobile device when the mobile device detects a match between the positioning data and the location data.

13. The inspection method of the building structures as claimed in claim 12, wherein when the mobile device detects a mismatch between the positioning data and the location data, the mobile device generates a prompt message.

14. The inspection method of the building structures as claimed in claim 12, wherein when the mobile device detects a mismatch between the positioning data and the location data, the mobile device does not display the augmented reality element on the screen.

15. The inspection method of the building structures as claimed in claim 11, wherein the digital environment integration platform includes at least one field photo corresponding to the at least one structural part model; the inspection method of the building structures further comprises: sending, by the mobile device, a request command to the server, wherein the request command is a request to acquire the at least one field photo; acquiring the at least one field photo from the digital environment integration platform by the server based on the request command and sending the at least one field photo to the mobile device by the server; displaying the at least one field photo on the screen by the mobile device.

16. The inspection method of the building structures as claimed in claim 11, wherein the mobile

device has user data; when the mobile device edits the inspection form, the user data is automatically inputted into the inspection form.

17. The inspection method of the building structures as claimed in claim 11, further comprising: generating, by the mobile device, a prompt message when the mobile device detects a mismatch between a shape of a structural part in the environment image and a shape of the augmented reality element of the at least one structural part model.

18. The inspection method of the building structures as claimed in claim 11, wherein the at least one structural part model defines a primary inspection portion and a secondary inspection portion; a color of the primary inspection portion is different from a color of the secondary inspection portion; when the mobile device displays the augmented reality element on the screen, the primary inspection portion and the secondary inspection portion are displayed simultaneously.

19. The inspection method of the building structures as claimed in claim 11, wherein the at least one structural part model defines a primary inspection portion and a secondary inspection portion; when the mobile device displays the augmented reality element on the screen, the mobile device adjusts the primary inspection portion on the augmented reality element to be semi-transparent; the mobile device compares the semi-transparent primary inspection portion on the augmented reality element with a portion of a structural part in the environment image; the mobile device generates a prompt message when a mismatch between the semi-transparent primary inspection portion on the augmented reality element and the portion of the structural part in the environment image is detected.

20. The inspection method of the building structures as claimed in claim 11, wherein when the mobile device displays the augmented reality element on the screen, the mobile device adjusts a transparency of at least one portion of the augmented reality element.